

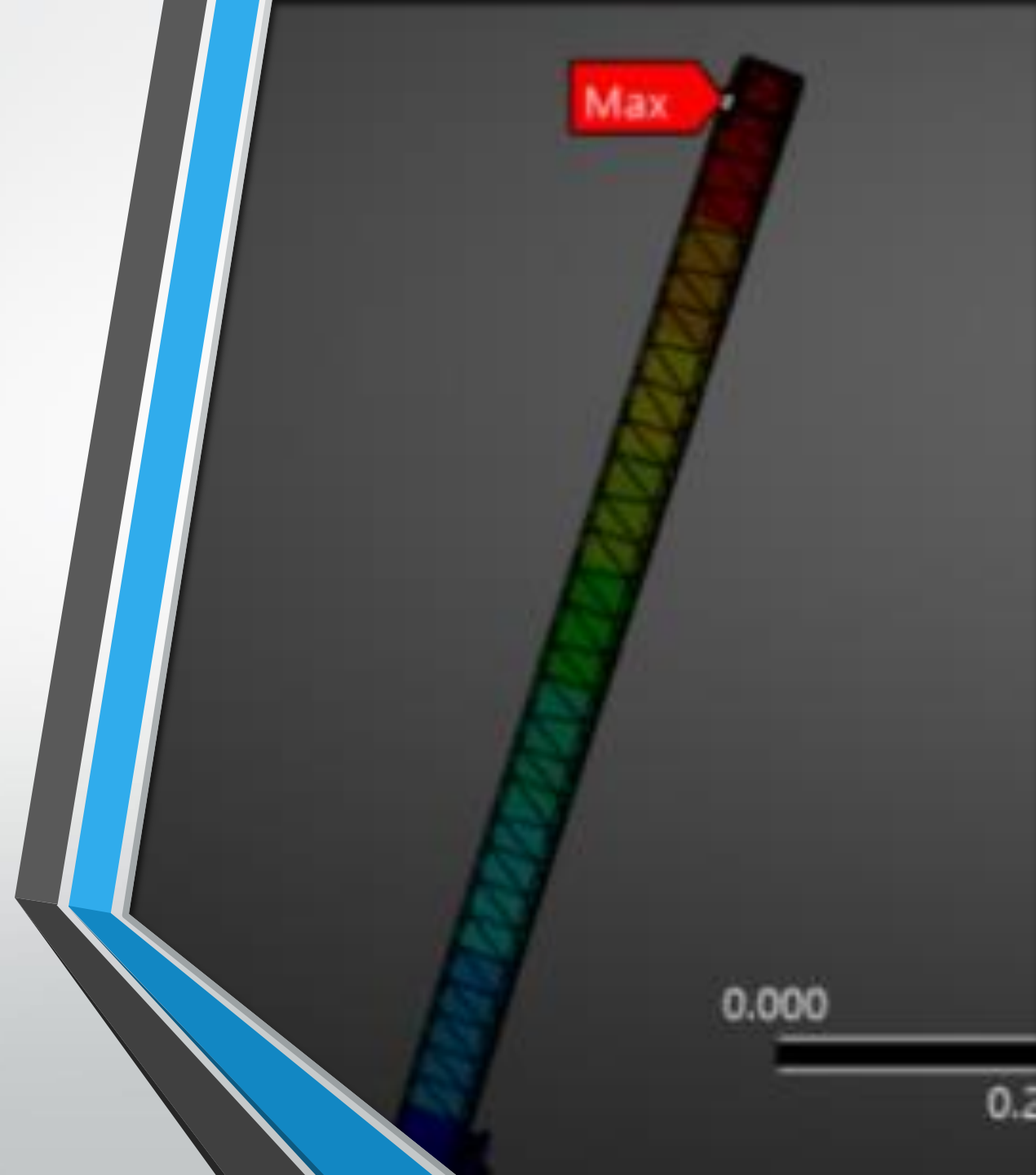


AIRFRAME TEAM

Week two Progress Report

1. Ansys simulation of the launchpad

- Using a mesh size of 6cm a maximum equivalent bending stress of 569.9 MPa and a deflection of 8.9mm in the force's direction were obtained.
- The video of the simulation is [here](#)
- **Purpose:** To simulate the various stresses(bending) occurring on the launchpad and oval deflection.

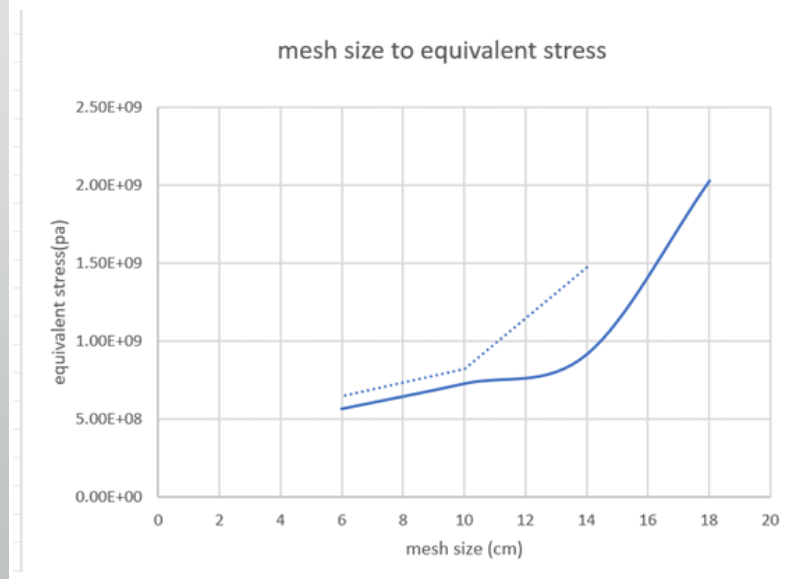


2.Mesh independence study of the launchpad

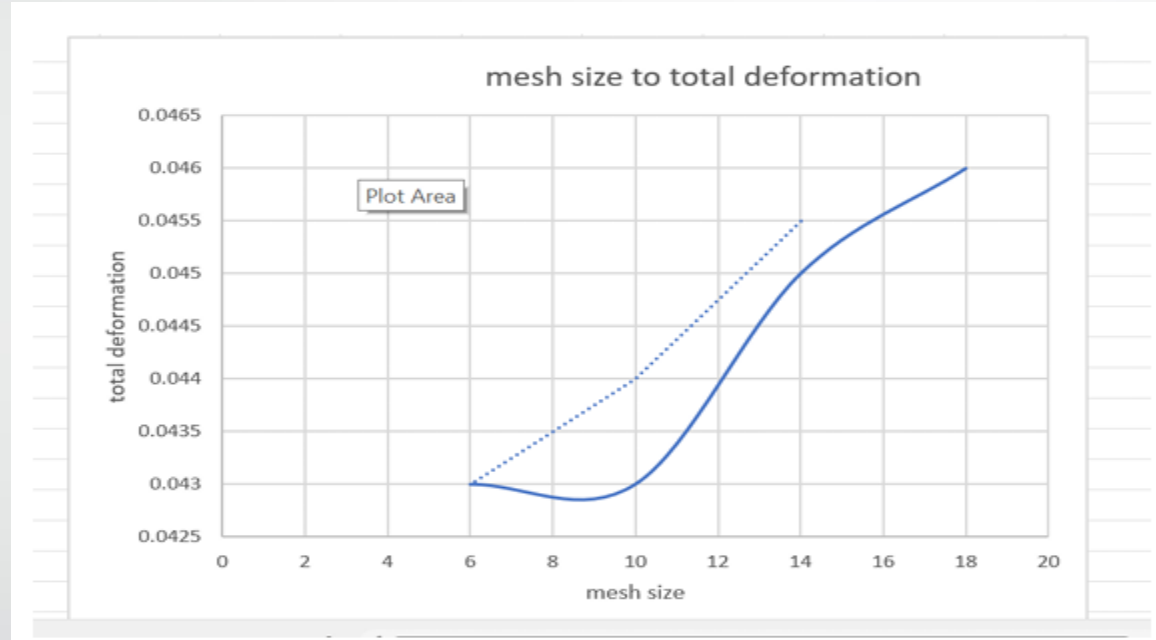
- The documentation is [here](#)
- The team investigated five mesh sizes (18cm, 14cm, 10cm, 6cm, 2cm). For each mesh size, the model was solved and the following table was obtained.
- The results indicate a significant reduction in predicted stress values as the mesh becomes finer. The stress values gradually approach a stable solution between the 12 cm and 2 cm mesh sizes, indicating convergence.

4.1 Mesh Statistics and Solution Results

Mesh Size (cm)	Nodes	Elements	Maximum Equivalent Stress (Pa)	Maximum Total Deformation (m)
18	92,514	44,220	2.03×10^9	0.046
14	100,043	48,047	9.19×10^8	0.045
10	111,362	53,677	7.31×10^8	0.043
6	126,144	60,160	5.699×10^8	0.043
2	177,122	84,172	5.78×10^8	0.043



- For deformation, the results remain nearly constant throughout the refinement process, demonstrating that deformation predictions are largely insensitive to further mesh refinement beyond the 6 cm mesh.



- Based on the convergence behavior observed during mesh refinement, the 6 cm mesh was selected for all subsequent analyses because it provides accurate results while minimizing computational requirements.

3. Theoretical calculations of stress(bending and axial), deflection, buckling analysis and safety of factor

- The team embarked on the theoretical analysis of the launchpad to verify the ansys simulations.
- The documentation is [here](#) and the following results were obtained:
- The deflection obtained is almost similar to the ansys simulation (0.78 of the simulation). The stresses can't be directly compared as the theoretical is directional while the simulation is equivalent.

Parameter	Value	Status / Notes
Bending stress σ_b	21.394 MPa	Well within yield (250 MPa)
Axial stress σ_{ax}	2.162 MPa	Low
Combined stress σ_c	23.556 MPa	Total compressive + bending
Tip deflection δ	1.1919 mm	serviceability OK
P_cr upper (K=2)	681.806 kN	Fixed-free overhang governs
P_cr lower (K=0.5)	2.727 kN	Fixed-fixed — higher capacity
SF_buckling	221.65	✓ Safe
SF_yield (combined)	10.61	✓ Safe
Governing failure mode	Bending / deflection	Not buckling or axial yield

4. Airframe Tasks preparation and listing

- The team was also able to prepare a mindmap of all the tasks that need to be undertaken [here](#)

5. Challenges experienced

- Lack of internet connection.
- Lack of access to workshops for fabrication.
- Computational resources required for Ansys.

Next week tasks

- Launchpad finishes: reaming, painting etc

THANK YOU

